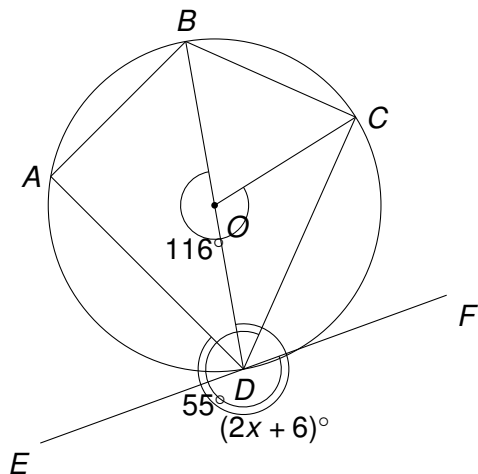


- 1 In the diagram, A , B , C and D are points on a circle with centre O . EDF is a tangent to the circle at D . BOD is a straight line (a diameter of the circle).

The angle at the centre, angle $BOC = 116^\circ$, angle $ADB = (2x + 6)^\circ$, and angle $CDF = 55^\circ$.



- (a) Find the size of angle BDC .

$BDC = \dots\dots\dots$
(2)

- (b) Work out the value of x .

$x = \dots\dots\dots$
(2)

(Total for Question 1 is 4 marks)
